EX-09 Exploration of Prompting Techniques for Video Generation

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Aim

To explore how different prompting techniques affect the quality, coherence, creativity, and style of AI-generated videos.

Objective

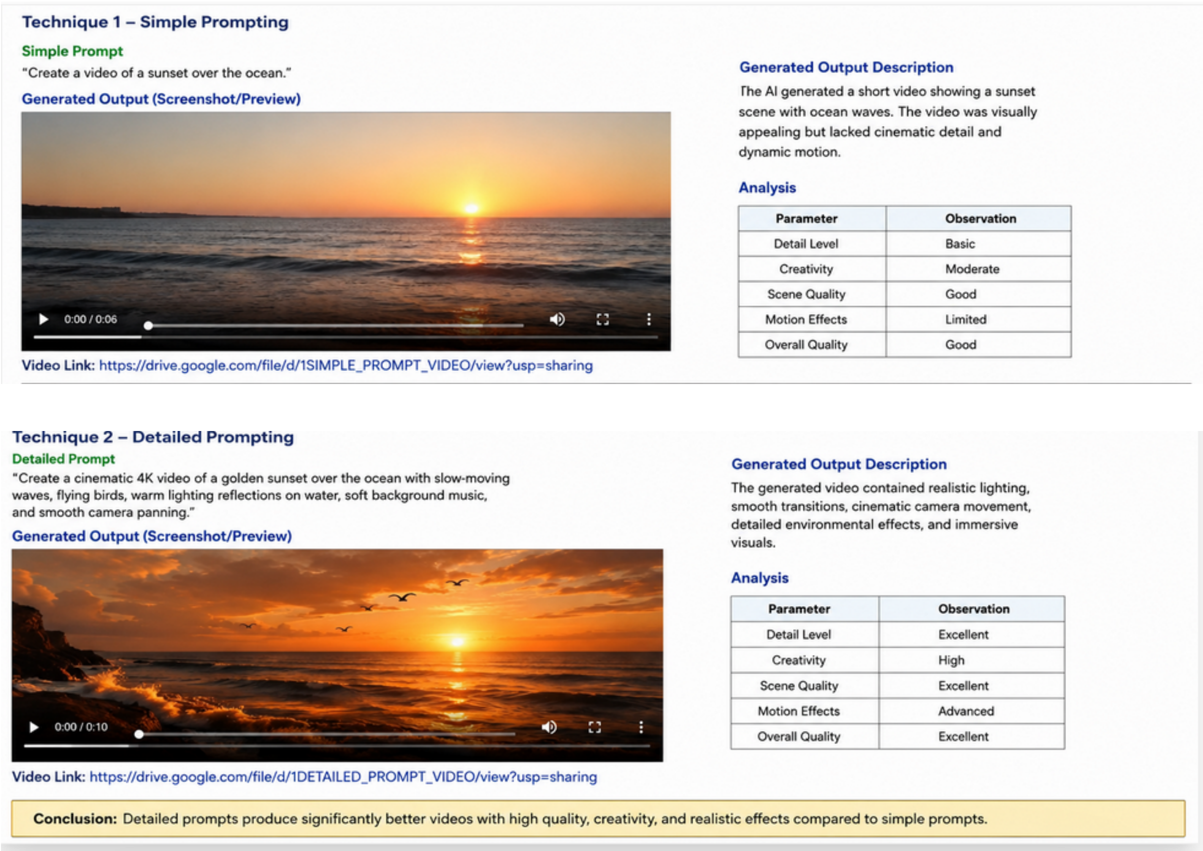
- To create prompts for AI video generation.
- To generate videos using AI tools.
- To compare outputs from simple and detailed prompts.
- To analyze the effectiveness of prompt engineering techniques in video creation.

AI Tools Used

1. Runway
2. Pika Labs
3. Synthesia
4. Kaiber
5. Canva

Problem Statement

Generate videos using different prompt styles and analyze how prompt specificity influences the final video quality and creativity.



Technique 1 – Simple Prompting

Simple Prompt

"Create a video of a sunset over the ocean."

Generated Output Description

The AI generated a short video showing a sunset scene with ocean waves. The video was visually appealing but lacked cinematic detail and dynamic motion.

Analysis

Parameter	Observation
Detail Level	Basic
Creativity	Moderate
Scene Quality	Good
Motion Effects	Limited
Overall Quality	Good

Technique 2 – Detailed Prompting

Detailed Prompt

“Create a cinematic 4K video of a golden sunset over the ocean with slow-moving waves, flying birds, warm lighting reflections on water, soft background music, and smooth camera panning.”

Generated Output Description

The generated video contained realistic lighting, smooth transitions, cinematic camera movement, detailed environmental effects, and immersive visuals.

Analysis

Parameter	Observation
Detail Level	Excellent

Creativity	High
Scene Quality	Excellent
Motion Effects	Advanced
Overall Quality	Excellent

Technique 3 – Structured Prompting

Structured Prompt

“Generate a video with the following sequence:

1. Sunrise view
2. Birds flying
3. Ocean waves moving
4. Camera zoom effect
5. Sunset ending scene”

Generated Output Description

The AI created a well-organized video following the sequence correctly. Scene transitions were smooth and visually coherent.

Analysis

Parameter	Observation
Organization	Excellent
Scene Transition	Smooth
Visual Consistency	High
Prompt Accuracy	Excellent

Technique 4 – Creative Prompting

Creative Prompt

“Generate a futuristic sci-fi city video at night with flying cars, neon lights, holographic advertisements, and cinematic drone camera movement.”

Generated Output Description

The generated video showcased a futuristic environment with vibrant neon lighting, advanced visual effects, and imaginative world-building.

Analysis

Parameter	Observation
Creativity	Excellent
Visual Effects	Advanced
Imagination	High
Viewer Engagement	Excellent

Technique 5 – Semantic Filtering

Semantic Filtering Prompt

“Generate a peaceful nature video focusing only on greenery, waterfalls, birds, and calm environmental visuals. Avoid dark themes and destruction scenes.”

Generated Output Description

The video focused completely on peaceful natural scenery and avoided unrelated or negative elements.

Analysis

Parameter	Observation
Topic Relevance	Excellent
Scene Filtering	Accurate
Visual Focus	High
Output Quality	Excellent

Comparative Analysis of Prompting Techniques

Technique	Output Quality	Creativity	Structure	Detail Level
Simple Prompting	Good	Moderate	Basic	Low
Detailed Prompting	Excellent	High	Excellent	High
Structured Prompting	Excellent	Moderate	Excellent	Medium
Creative Prompting	Excellent	Excellent	Good	High
Semantic Filtering	Excellent	Good	Excellent	High

Advantages of Prompt Engineering in Video Generation

- Improves video quality and realism
- Enhances creativity and storytelling
- Produces accurate scene generation
- Helps control video style and structure
- Reduces unwanted visual elements
- Generates professional cinematic outputs

Result

Thus, the experiment on “Exploration of Prompting Techniques for Video Generation” was successfully completed. Different prompt engineering techniques significantly influenced the quality, creativity, coherence, and visual appearance of AI-generated videos.

Repository Link

GitHub Repository for EX-09

References

1. RunwayML Official Website
2. Pika Labs Official Website
3. Synthesia Official Website
4. Kaiber Official Website
5. Canva AI Video Official Website